

AeroPulse-NG: Offline Tactical Radar & Airspace Surveillance

Entry category: Flight Planning & Air Traffic Management · Team lead submission via official NDAIE form · Word count 285/300 (rulebook §3).

Problem

Nigeria's airspace is monitored largely through expensive imported primary and secondary surveillance radar — installations costing millions of dollars, demanding specialist maintenance, yet still leaving low-altitude corridors, secondary aerodromes and terrain-shadowed sectors under-covered. When connectivity or a radar head fails, controllers lose the traffic picture precisely when harmattan haze, diversion surges or military activity raise risk. Non-cooperative and transponder-silent aircraft widen the gap further.

Proposed solution

AeroPulse-NG is an air-gapped surveillance engine that turns commodity software-defined radios (about USD 150 of hardware) plus a standard PC into a working tactical radar display. It ingests over-the-air aviation telemetry with zero internet or cloud dependency, giving civil air traffic management (NAMA) and tactical command (NAF) a sovereign, locally maintainable traffic picture at a fraction of conventional cost.

Technical approach

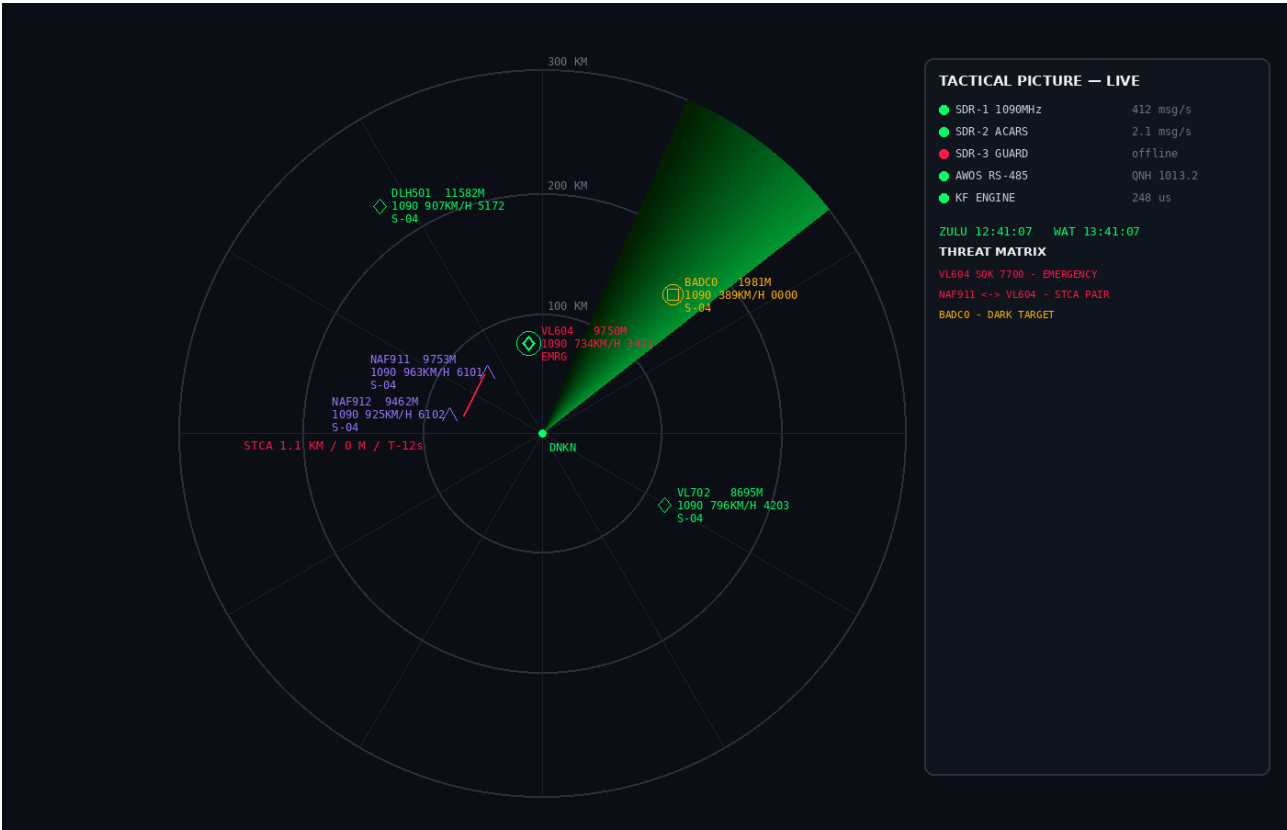
A Rust/Tauri backend demodulates 1090 MHz Mode S ADS-B, resolving positions through Compact Position Reporting; a six-state Kalman filter (constant-velocity, linear; module `ekf.rs` historic) smooths jitter and dead-reckons tracks through 30-second RF dropouts. An R*-tree spatial index drives continuous Short-Term Conflict Alerting against ICAO Doc 4444 separation minima (rendered 9.26 km / 305 m) within a 120-second lookahead. A triple-fusion weather matrix merges AWOS serial telemetry, ACARS D-ATIS broadcasts and Mode S BDS 4,4/4,5 downlinks into an offline 3D wind/temperature model tuned for harmattan operations. Defence overlays flag dark targets, squawk emergencies and geofence incursions, and compute intercept geometry. A hardware-accelerated display layer renders HF-STD-010A-compliant screens across dual operator monitors, in SI metric units per NCAA / ICAO Annex 5 policy.

Likely impact

Operational: conflict alerting and fused weather at aerodromes that cannot justify radar. Economic: over 99% capital reduction per station. Sovereignty: fully offline national capability. Educational: an open engineering reference for Nigerian avionics talent. Development stage: prototype core (decoder, Kalman filter, STCA engines) verified by 129 automated checks; dual-display operator interface complete; RF front-end and field-trial hardware integration next.

Verification Snapshot

Suite	Checks	Covers
Rust engine (cargo)	108	Decoders, CPR, Kalman filter, R*-tree, STCA, fusion pipeline end-to-end
Python sidecar (pytest)	21	AWOS codec, METAR parser, DuckDB durability
TypeScript strict (tsc)	clean	Wire contracts, SI unit policy, components
Production bundle (vite)	build	Dual-window workspace, 65 KB gzipped



Demonstration scenario

- T+45 s — VL604 squawks 7700 (general emergency) with TC-28 status frame.
- T+70 s — NAF911 intercept crosses VL604's path: predicted miss under 1 km triggers STCA.
- T+112 s — VL604 enters RF dropout: Kalman filter dead-reckoning keeps the picture alive.
- Rolling — dark target flagged, ACARS METAR and AWOS frames feeding the fusion matrix.